Meijie Wang

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Education

Xiamen University — *Ph.D. in Condensed Matter Physics, GPA 3.91* (Expected) Sep 2022 – Jun 2028

- Focus: computational materials, single/dual-atom catalyst design.

Hangzhou Normal University — *B.S. in Physics* Sep 2018 – Jun 2022

Work Experience

Deep Principle — *AI4S Algorithm Intern* Jan 2026 – Present

- Trained and evaluated MPA, a materials foundation model bringing LLM-style multi-phase training (pre-/mid-/post-training) to experimental property prediction — **SOTA on 35/40 tasks** and **14.6% lower MAE** than direct fine-tuning under scaffold (OOD) splits, surpassing Uni-Mol2, Suiren, and ChemProp.
- Implemented the **mid-training** (physics-guided alignment on large-scale first-principles data) and **post-training** pipelines — including a **Hybrid Readout** head (attention-pooling + atom-additive) — built the training/evaluation infrastructure and ran all large-scale training.

Xiamen University — *Ph.D. Research* Sep 2022 – Present

- **Dual-atom catalyst design & mechanism:** designed Si-based dual-atom catalysts for CO_2 reduction and uncovered the p–d orbital-coupling mechanism; screened **360+ candidates** with a GBR pipeline. (Appl. Surf. Sci. 2024, J. Mater. Chem. A 2024)
- **Curvature-driven catalysis:** established curvature as an independent activity knob (**inverted-volcano** relation, interpretable descriptor $\mathbf{R^2 = 0.92}$) and generalized it into a unified **geometric-electronic principle**. (J. Phys. Chem. Lett. 2026, ACS Catal. 2026)

Selected Publications & Patents

* first-author

Report: **Materials Property Axiom: Adapting Foundation Models to Experimental Property Prediction via Multi-phase Training**
Deep Principle Team, **Technical Report, 2026**

Paper*: **A Geometric-Electronic Principle for Curvature-Driven Catalysis**
M. Wang, Y. Lin, Z. Huang, Y. Sun, Z.-Z. Zhu, S. Wu, X. Cao, **ACS Catal., 2026 (Accepted)**

Paper*: **Curvature Engineering of SiFe Dual-Atom Catalysts for Enhanced CO_2 Electroreduction**
M. Wang, Y. Lin, Y. Xiang, Y. Sun, Z.-Z. Zhu, S. Wu, X. Cao, **J. Phys. Chem. Lett., 17, 1227–1234 (2026)**

Paper*: **p–d Orbital Coupling in Silicon-Based Dual-Atom Catalysts for Enhanced CO_2 Reduction**
M. Wang, Y. Xiang, Y. Lin, Y. Sun, Z.-Z. Zhu, S. Wu, X. Cao, **J. Mater. Chem. A, 12(46), 31902–31913 (2024)**

Paper*: **$SiFeN_6$-graphene: A Promising Dual-Atom Catalyst for Enhanced CO_2-to-CH_4 Conversion**
M. Wang, Y. Xiang, W. Chen, S. Wu, Z.-Z. Zhu, X. Cao, **Appl. Surf. Sci., 643, 158724 (2024)**

Patent: **A Liquid Detection Device**
M. Wang, R. Qin, S. Zhou, W. Ding, Y. Che, **China Invention Patent ZL 2021 1 0042048.9, granted 2024.05**

Skills

- **Foundation Models:** Foundation models, pre-/mid-/post-training, custom readout heads and loss design, PEFT/LoRA fine-tuning, equivariant GNN potentials.
- **ML Engineering & Evaluation:** Training/evaluation infrastructure, distributed training (DDP), data curation, benchmark design, statistical evaluation.
- **Programming & Computation:** Python, PyTorch, NumPy/Pandas/Matplotlib, Linux, HPC/SLURM, DFT (VASP, JDFTx), ASE, pymatgen, scikit-learn.